
When Shift Happens - Confounding Is to Blame

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Abstract

Distribution shifts directly threaten the robustness and generalization of machine learning models. Although the dominant view holds that causal-invariant representations are the reliable route to robustness, recent empirical evidence establishes a sharper puzzle: (i) empirical risk minimization (ERM) can match or surpass state-of-the-art out-of-distribution (OOD) generalization methods, and (ii) ERM improves when it uses all available covariates rather than restricting itself to causal ones. We identify hidden confounding as the mechanism behind this behavior. Shifts in hidden confounders generate distribution changes that break the assumptions underlying many OOD generalization methods. Under these conditions, we prove that strong generalization requires environment-specific relationships in addition to invariant ones. We further show that proxy variables for hidden confounders directly alleviate the resulting generalization challenge. Together, the theory and experiments provide a decisive explanation for the empirical strength of ERM and deliver practical guidance for robust OOD algorithms and covariate selection.

1 Introduction

Generalization—the ability to draw reliable conclusions about unseen data from observed data—is a core requirement across scientific disciplines. In medicine and the social sciences, it appears as *external validity*, which demands that findings from one population transfer to another [13]; in ecology, it underwrites *space-for-time substitutions*, where spatial variation supports inferences about long-term temporal change [60]; and in engineering, it anchors *robust control*, where systems must retain performance under unmodeled disturbances [35]. Machine learning now plays a central role in this agenda by providing models that are expected to generalize beyond the data used to train them [100, 88, 49].

Robustness to distribution shift is therefore not optional: it is the central test of whether a learned model generalizes. A model trained to predict *food stamp reciprocity* from *household attributes* in one state, for example, must remain reliable when deployed in another state. We say that a model achieves *out-of-distribution (OOD) generalization* when it preserves performance on both in-distribution (ID) data, from which training data are sampled, and OOD test data. Over the last decade, domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94] have produced methods with provable guarantees under specific assumptions. Yet careful model selection [30, 38, 87] repeatedly shows that standard *empirical risk minimization (ERM)* [86] remains a formidable baseline and often achieves competitive OOD performance in real applications [30, 24, 50, 57, 67, 87]. Even more pointedly, a recent study on 16 real-world tabular datasets finds that using all available covariates, rather than only those judged causal, can improve OOD generalization [57]. These results do more than expose a benchmarking anomaly; they reveal a gap in the standard explanation of OOD generalization.

Distribution shifts naturally arise when data come from different environments [71]. For example, *food stamp reciprocity* can vary across states because eligibility rules differ, producing distinct environment-specific distributions. A major line of OOD work seeks statistical relationships between covariates and outcomes that remain stable across environments, known as *invariant relationships* [6]. Learning such relationships can support OOD generalization under the assumptions used by invariant-learning methods [6, 59, 31, 66, 56, 98, 63]; causal invariance is the important special case in

which the causal relationships themselves remain stable [71, 59, 6]. Real tabular prediction problems, however, rarely expose all variables needed to sustain these invariances. Data-collection limits, privacy constraints, and measurement errors routinely leave relevant covariates unobserved [14, 52, 20, 48]. When the missing variables are confounders that affect both observed covariates and the outcome, their distributional shifts define distinct environments and directly distort observed relationships. Hidden confounding is therefore not a peripheral nuisance: it is a primary mechanism that degrades generalization [40, 4, 85, 62] and can force models to learn relationships with the wrong sign.

The role of hidden confounding shifts in OOD generalization has received far less attention than its importance warrants [4, 40, 85, 62]. This paper closes that gap with a theory and empirical analysis that explain why ERM and all-covariate models can succeed under real-world shifts [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- We establish hidden confounding shift as a central OOD generalization problem from a causal perspective (§ 3) and explain why informative, non-causal covariates can improve performance.
- We prove that maximizing predictive information between model predictions and true outcomes requires learning environment-specific relationships. Invariant representations remain valuable, but they are not sufficient on their own (§ 4.1, § 4.2).
- We show that variables informative about either outcomes or hidden confounders improve predictive information, making principled covariate selection essential under hidden confounding shift (§ 4.3).
- Across real-world and synthetic datasets, we demonstrate that (i) hidden confounding signatures are widespread in tabular benchmarks, (ii) environment-specific relationships track both ID and OOD accuracy, and (iii) informative non-causal covariates improve generalization (§ 5).

Overall, the paper gives a direct explanation for the empirical strength of ERM under real-world distribution shifts and positions hidden confounding as an unavoidable driver of OOD behavior that future methods must address explicitly.

2 Related work

OOD generalization spans several related settings, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. Domain adaptation uses unlabeled test-domain data, domain generalization uses environment labels during training, federated learning coordinates multi-environment learning under communication and privacy constraints, and OOD detection identifies samples that depart from the training distribution. Across these areas, invariant relationships are a dominant design principle. Recent work, however, makes clear that invariance alone does not explain or deliver the strongest generalization [46, 72, 93]. In particular, Ye et al. [93] show that feature invariance must be paired with feature *informativeness*. Our analysis puts these two requirements in one decomposition and specifies how hidden confounding changes their roles.

Proxy variable adjustment: When confounding variables are observed at training time but absent at test time, Landeiro and Culotta [40] adjust for confounding shifts to improve classifier performance. Proxy-based causal effect estimation [55, 39] has also motivated recent domain adaptation methods that cast unknown distribution shift as causal effect identification [4, 85]. OOD generalization under hidden confounding shift has further been studied under overlapping confounder support [62]. Our contribution is distinct: we explain, in predictive-information terms, exactly why proxy variables improve OOD generalization.

All variable models vs causal models. Nastl and Hardt [57] provide a benchmark in which covariates are divided into causal, arguably causal, anti-causal, and other spurious groups. Across 16 datasets, models using all covariates Pareto-dominate models restricted to causal or arguably causal subsets on both ID and OOD data. The empirical pattern is clear, but its theoretical explanation has been limited; our hidden-confounding analysis supplies that explanation. For linear causal models, anchor regression [68] balances between using all observed covariates and focusing only on causal covariates. We address the complementary question of why adding non-causal covariates can be beneficial under hidden confounding shift. Relatedly, Eastwood et al. [21] show that unstable covariates can improve performance when they carry label information and are conditionally independent of stable covariates given the label, and they adjust shifts using unlabeled test-domain data. Yet on a medical real-world dataset not constructed for that specific setting [8], ERM remains competitive, matching the broader

findings of Nastl and Hardt [57]. This is consistent with the fact that, under well-specified covariate shifts, maximum likelihood estimation (MLE) is minimax optimal for OOD generalization [26]. Real-world shifts, however, are rarely that well-specified: hidden confounding is precisely the regime where the all-covariate advantage becomes theoretically meaningful.

3 Manifestations of hidden confounding shift

This section makes hidden confounding shift concrete and shows why OOD methods must confront it directly, using causal effect identification for observed covariates as the guiding example.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , Table 1 lists the distribution shifts that can separate two environments e and e' . These shifts are naturally produced by changes in the distribution $\mathbb{P}(U)$ of an unobserved variable U that causes $\mathbf{X}$, Y , or both (Figure 1). A shift in $\mathbb{P}(U)$ becomes visible through observed distributions involving only $\mathbf{X}$ and Y . For example, $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$, so $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ together with $U \rightarrow \mathbf{X}$ directly implies $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The same logic applies to shifts in $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, making hidden confounding a single mechanism that can manifest as multiple familiar shift types.

Table 1: Core distribution shifts induced by environment changes.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

Many OOD methods derive their guarantees from causal relationships among U , $\mathbf{X}$, Y that enforce particular invariances. For instance, when $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), since $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$, a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to label shift, i.e., $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, but conditional covariate distribution stays invariant i.e., $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ because $\mathbb{P}(\mathbf{X}|Y, U) = \mathbb{P}(\mathbf{X}|Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Similarly, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to covariate shift, i.e., $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, but the conditional distribution $\mathbb{P}(Y | \mathbf{X})$ stays invariant i.e., $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These invariances may not hold in many scenarios because U usually causes both $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In this case, when $\mathbb{P}(U)$ shifts between environments, label shift, covariate shift, conditional covariate shift, and concept shift can all occur simultaneously. Nevertheless, confounding shift induces an invariance: $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$ referred to as *stable confounding shift* [85, 4]. Since U is unobserved, this invariance is absent from the observed data.

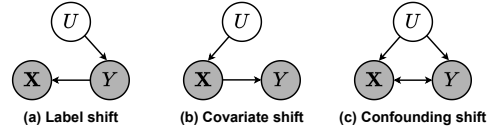


Figure 1: Causal graphs that generate the observed distribution shifts.

Information-theoretic measures: Following Federici et al. [22], we quantify shifts with mutual information. The quantities $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, $I(\mathbf{X}, Y; E)$, where E denotes the environment, measure shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$ respectively. This notation makes the key conflict explicit: label-shift methods often assume $I(\mathbf{X}; E | Y) = 0$, while covariate-shift methods often assume $I(Y; E | \mathbf{X}) = 0$. Hidden confounding can make both quantities nonzero at the same time.

Challenges with hidden confounding shift: To illustrate the challenges posed by hidden confounding shift on OOD generalization, we present the experiment shown in Figure 2. The outcome Y depends on X and a hidden confounder U . Environment-specific shifts in the mean of U , denoted by μ_e , induce distinct environments. A linear regression model trained solely on observed covariate X infers an incorrect relationship between X and Y , as shown in the first subplot. This exemplifies the Simpson’s paradox [73], where Y increases with X within each environment, yet the model captures the opposite trend. We observe similar behavior from the models designed for OOD generalization, such as IRM [6], VREX [38], and Group DRO [69]. In such scenarios, we theoretically show in § 4 that the optimal strategy for generalization involves learning environment-specific relationships, which enables recovery of the correct relationship between X and Y . This is illustrated in the second subplot, where environment-specific summary statistics of X , such as mean, standard deviation, and quantiles, help uncover the true X – Y relationship. This mirrors the *backdoor adjustment* criteria in causal effect estimation [58], where environment-specific information acts as a proxy for adjusting for

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

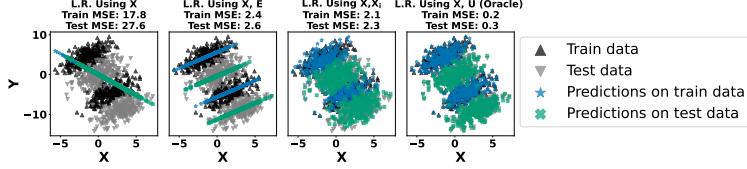


Figure 2: Four linear regression (L.R.) models expose the effect of hidden confounding shifts under minimal environment overlap (i.e., distant μ_e). (i) Training only on X learns the wrong relationship with Y , exhibiting Simpson’s paradox. (ii) Environment-specific summary statistics of X , denoted as E , recover the correct relationship but have limited representational power. (iii) An informative covariate X_i for U improves OOD generalization. (iv) The oracle model uses both X and U .

confounders. However, as shown in the second subplot, environment-specific information encoded in observed covariates alone may be insufficient due to the limited representational capacity. However, as we show in § 4.3, additional informative covariates serving as proxies for unobserved confounders can further improve generalization performance, as shown in the third subplot. For comparison, we include an oracle model trained on X and U , which achieves the best-possible mean squared error.

4 Impact of hidden confounding shifts on OOD generalization

This section develops the theoretical account of how hidden confounding shifts reshape OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ be the predicted label for an input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation function. $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ denotes entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ denotes mutual information. The risk of the predictor $(f \circ \phi)$ in an environment e is defined as: $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X},Y) \sim \mathbb{P}_{\mathbf{X},Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ where $\ell(\cdot, \cdot)$ is a loss function. The goal in OOD generalization is to learn a predictor $(f \circ \phi)$ that performs well on both ID and OOD data. To this end, various objectives have been considered in the literature. Robust optimization-based methods [7, 9, 19, 75, 69] aim to minimize the worst risk across all training environments: $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ where $\mathcal{E}_{tr}$ denotes the set of training environments. The variance risk extrapolation (VREX) [38] aims to minimize $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$ where β is a hyperparameter. Invariant risk minimization (IRM) [6] aims to minimize $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ with the constraint that f is a simultaneously optimal classifier across all environments. Empirical risk minimization (ERM) based methods simply pool data from all training environments and minimize the empirical risk on the pooled data [6, 38]. That is, ERM minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X},Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$ where $\mathcal{D}^e$ denotes observed data in an environment e .

To expose how hidden confounding shift acts on standard objectives, we analyze predictive information, $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$. This quantity directly measures how much the prediction reveals about the true label, so it provides a precise objective for studying generalization. We then isolate two properties of $\phi(\mathbf{X})$ —*informativeness* and *invariance*—that are central to OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness captures the label-relevant information that $\phi(\mathbf{X})$ retains within each environment. Reducing CINF necessarily discards predictive signal. Maximizing CINF can be inappropriate in fairness settings because $\phi(\mathbf{X})$ may exploit sensitive or biased within-environment information, and reducing such bias by minimizing $I(\phi(\mathbf{X}); E)$ can lower predictive performance, as shown in § 4.2. In the setting studied here, where E is not treated as sensitive information, maximizing CINF is a valid and powerful route to stronger generalization; § 4.1 makes this claim formal.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$.*

193 Invariance measures whether $\phi(\mathbf{X})$ remains stable across environments for a fixed label Y . Low
 194 invariance preserves environment-specific signal and can overfit to training environments, whereas
 195 high invariance removes such dependencies and supports invariant learning. Building on the shift
 196 measures in § 3, we use $I(\phi(\mathbf{X}); E)$ for *feature shift* and $I(Y; E | \phi(\mathbf{X}))$ for *concept shift*. Depending
 197 on context, *concept shift* denotes either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

198 4.1 A general decomposition of predictive information

199 We begin with a few causal-graph preliminaries. A causal graph $\mathcal{G}$ consists of nodes representing
 200 random variables, and directed edges indicating direct causal influences between nodes. A *path*
 201 between two nodes X_i and X_j is a sequence of unique nodes connected by edges. A *directed path*
 202 from X_i to X_j with $i < j$ is one where all arrows point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow$
 203 $X_{j-1} \rightarrow X_j$. In such a directed path, X_i is called the *parent* of X_{i+1} , X_{i+1} is the *child* of X_i , X_i an
 204 *ancestor* of X_j , and X_j is a *descendant* of X_i . Paths decompose into three fundamental structures: a
 205 *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In both chains and
 206 forks, X_i and X_k are marginally dependent yet become conditionally independent upon conditioning
 207 on the intermediate node X_j , i.e. $X_i \not\perp X_k$ and $X_i \perp X_k | X_j$. In a collider, X_i and X_k are
 208 marginally independent but become conditionally dependent when conditioning on X_j or any of its
 209 descendants, i.e. $X_i \perp X_k$ and $X_i \not\perp X_k | X_j$. A path between X_i and X_k is said to be *blocked* by
 210 a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies
 211 in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$.
 212 If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp X_k | \mathcal{S}$. A path is open if it is not blocked.

213 Now consider the predictive information $I(Y; \hat{Y})$, where the
 214 predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ are based on a learned representa-
 215 tions $\phi(\mathbf{X})$. We model the underlying causal relationships
 216 among $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$ and E as shown in Figure 3. Here,
 217 U is a hidden confounding variable. Given either $\mathbf{X}$ or $\phi(\mathbf{X})$,
 218 $\hat{Y}$ is redundant for reasoning about Y . That is, $\hat{Y} \perp Y | \mathbf{X}$
 219 and $\hat{Y} \perp Y | \phi(\mathbf{X})$. E denotes an environment variable that
 220 captures shifts in $\mathbb{P}(U)$. That is, between any two environments,
 221 there is a shift in $\mathbb{P}(U)$. We first present a more general decom-
 222 position of predictive information without explicit consideration of how U influences $\mathbf{X}, Y$. A formal
 223 treatment of U 's influence on $\mathbf{X}$ and Y is presented in § 4.2.

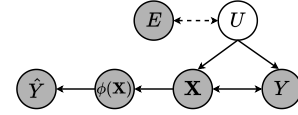


Figure 3: The bi-directed arrow between $\mathbf{X}$ and Y marks the mixed causal direction: some covariates in $\mathbf{X}$ can cause Y , while others may be caused by Y . A formal treatment of U 's influence on $\mathbf{X}$ and Y is presented in § 4.2.

224 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 225 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 226 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y | E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E | Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E | \phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y | \hat{Y})}^{\text{Residual}} \quad (1)$$

227 where $I(\phi(\mathbf{X}); Y | \hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 228 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

229 Appendix § A gives the proofs. Equation (1) provides a unifying lens: existing OOD methods can be
 230 interpreted by the decomposition terms they explicitly or implicitly optimize. For instance, IRM [6]
 231 enforces that a fixed classifier f remains the same across different environments. This is directly
 232 related to maximizing $-I(\phi(\mathbf{X}); E | Y)$ because if $\phi(\mathbf{X})$ were to carry extra environment-specific
 233 information conditional on Y , the fixed classifier would no longer be optimal in all environments.
 234 DANN [23] and the independence criterion for fair classification [22] aims to minimize environment-
 235 specific information in $\phi(\mathbf{X})$ by minimizing $I(\phi(\mathbf{X}); E)$. In the context of domain adaptation, certain
 236 properties of features are learned invariant to domains [82, 90]. While minimizing $I(\phi(\mathbf{X}); E)$ may
 237 be good for fair classification, it may degrade predictive performance [98, 33, 22]. This can be seen
 238 from Equation (1), where minimizing $I(\phi(\mathbf{X}); E)$ can reduce $I(Y; \hat{Y})$. In CDAN [51], the objective
 239 is to obtain $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^e(\phi(\mathbf{X}), Y)$. By enforcing this equality, E becomes independent of
 240 the pair $(\phi(\mathbf{X}), Y)$. Consequently, $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$.
 241 Thus, although CDAN does not explicitly include $I(Y; E | \phi(\mathbf{X}))$ in its loss, its joint-distribution
 242 alignment objective effectively drives this conditional mutual information to zero. While $I(Y; E)$ is
 243 a constant, there exist methods that are robust to shifts in the distribution of labels [69, 95].

Equation (1) deliberately abstracts away from the precise causal role of U . Once that role is specified, the decomposition identifies which terms should be targeted. If $U \rightarrow Y, U \not\rightarrow X, Y \rightarrow X$, then $I(\mathbf{X}; E | Y) = 0$, making minimization of $I(\phi(\mathbf{X}); E | Y)$ a principled objective. If $U \not\rightarrow Y, U \rightarrow X, X \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, which motivates minimizing $I(Y; E | \phi(\mathbf{X}))$. The genuinely difficult regime is hidden confounding, where $U \rightarrow Y$ and $U \rightarrow X$. There, the decomposition terms interact nontrivially, and the next section identifies the appropriate strategy.

4.2 Predictive information under hidden confounding shift

To understand how hidden confounding variable U that cause both X and Y , impact the predictive information, we consider *two special cases* of the relationship between X and Y : (i) $X \rightarrow Y$ and (ii) $Y \rightarrow X$. When $X \rightarrow Y$, we obtain the inequalities in (2) because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow X \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider node Y which results in additional information flow from $\phi(\mathbf{X})$ to E via Y . In contrast, conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the node X as long as $\phi(\mathbf{X})$ encodes some information about X .

$$(i) \overbrace{I(\phi(\mathbf{X}); E | Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E | \phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Similarly, when $Y \rightarrow X$, we obtain the inequalities in (3) because conditioning on Y blocks the information flow from $\phi(\mathbf{X})$ to E at the node Y and conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow X \leftarrow U \leftarrow E$ from Y to E at the node X because $\phi(\mathbf{X})$ is the child of X and conditioning on the child of a collider opens a path via that collider [58] (recall the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E | Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E | \phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Using the inequalities in (2), (3), we refine the predictive information decomposition as follows.

Proposition 4.2. *For a covariate vector X , label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both X and Y , and if either (i) $X \rightarrow Y$ or (ii) $Y \rightarrow X$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y | E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y | \hat{Y})}^{\text{Residual}} \quad (4)$$

Proposition 4.2 handles $X \rightarrow Y$ and $Y \rightarrow X$ separately, but its most common empirical use case is $X \rightarrow Y$. In recent tabular benchmarks, 11 of 16 datasets follow the causal structure $X \rightarrow Y$ [57, 50, 24]; in the remaining 5 datasets, covariates caused by Y are fewer than covariates that cause Y . The central implication of the proposition is therefore broadly relevant:

Compared with (1), the hidden-confounding decomposition in (4) for (i) $U \rightarrow X, U \rightarrow Y, X \rightarrow Y$ and (ii) $U \rightarrow X, U \rightarrow Y, Y \rightarrow X$ eliminates *variation, label shift, feature shift, and concept shift* as separate optimization targets. Thus, under hidden confounding shift with either $X \rightarrow Y$ or $Y \rightarrow X$, maximizing predictive information reduces to maximizing the gap *conditional informativeness* – *residual*.

Maximizing conditional informativeness: Prashant et al. [62] recently proposed a mixture-of-experts (MoE) model for OOD generalization under hidden confounding shift, with each expert tied to a hidden-confounder assignment. Each expert therefore maximizes performance within the environment associated with a particular confounder value, exactly matching the goal of maximizing $I(\phi(\mathbf{X}); Y | E)$. Equation (4) gives a direct theoretical justification for MoE-type models and, more importantly, shows what remains unresolved. Current methods such as Prashant et al. [62] assume overlapping confounder support, $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$, and discrete proxy variables; the decomposition calls for methods that work beyond those restrictions.

Figure 2 demonstrates that even without confounder-support overlap and with a continuous-valued proxy, a linear regression model can recover the correct causal relationship between observed covariates and the target when it receives environment-specific information. This example makes the algorithmic requirement concrete: OOD methods should remain effective under limited confounder overlap and continuous confounding structure.

279 **Minimizing residual:** The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ quantifies how much additional information
 280 $\phi(\mathbf{X})$ provides about the true label Y beyond what is already contained in the prediction $\hat{Y}$. The
 281 residual term can be expressed as: $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional
 282 entropy $H(Y | \hat{Y})$ is related to the expected cross-entropy loss as: $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y |$
 283 $\hat{Y}) + \text{calibration error}$ [12, 10, 29]. Here $H(Y | \hat{Y})$ is known as the *refinement error* that measures
 284 the model’s ability to distinguish between classes. Minimizing the residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ thus
 285 involves two complementary strategies. First, one can directly minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ by making
 286 $\phi(\mathbf{X})$ conditionally independent of the target Y given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can
 287 minimize the refinement error $H(Y | \hat{Y})$, for instance, through training techniques that minimize
 288 refinement error for accurate predictions before minimizing the calibration error [10].

289 4.3 Impact of additional informative variables

290 In § 4.2, we present the optimal strategies for generalization when the confounder is unobserved.
 291 In practice, we sometimes have access to proxies for the hidden confounding variable [4, 85, 62],
 292 which can be leveraged to substitute for the hidden confounding variable. We now study the impact
 293 of adding such non-causal but informative covariates on predictive information.

294 **Definition 4.3** (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e.,*
 295 *neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are*
 296 *not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

297 Since $U \rightarrow Y$, any covariate that is informative to U , is also informative to Y and vice-versa. From
 298 the predictive information decomposition in (1), to maximize $I(Y; \hat{Y})$, it is required to minimize the
 299 concept shift $I(Y; E | \phi(\mathbf{X}))$. Liu et al. [50] suggest based on their empirical analysis that collecting
 300 additional covariates $\mathbf{X}_I$ such that $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ is more stable across environments i.e., reducing
 301 concept shift, improves the OOD test accuracy. We theoretically show that utilizing more informative
 302 variables helps maximizing predictive information by maximizing or minimizing certain terms in (1).

303 **Proposition 4.3.** *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that*
 304 *$Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} & (i) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y | E) > I(\phi_1(\mathbf{X}); Y | E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ & (iii) I(Y; E | \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E | \phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

305 Proposition 4.3 establishes the precise effect of informative covariates: they increase conditional
 306 informativeness (5.i) and feature shift (5.ii), while decreasing concept shift (5.iii). The same
 307 covariates also increase variation (5.iv), so strong methods should exploit their informativeness while
 308 explicitly controlling the added variation.

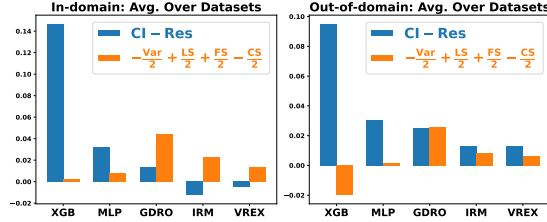
309 5 Experimental results

310 Our experiments directly test the theoretical claims on real-world and synthetic datasets: (i) hidden
 311 confounding shift is present in real-world data, (ii) the components of (1) explain OOD generalization
 312 under hidden confounding shifts, and (iii) informative covariates improve generalization. We consider
 313 eight real-world tabular benchmark datasets: *Food stamps, Readmission, Income, Public coverage,*
 314 *Unemployment, Diabetes, Hypertension, and ASSISTments*. These datasets and corresponding
 315 domain splits are adopted from *TableShift* benchmark [24]. We use ID test and OOD test accuracies to
 316 measure the performance of models. We perform experiments on two ERM-based methods: XGBoost
 317 (XGB) [15] and multi-layer perceptron (MLP), two domain generalization methods: IRM [6],
 318 VREX [38], and one robust learning based method: Group DRO (GDRO) [69]. We present average
 319 results over 5 different random hyperparameter choices as described in Appendix B. The effect of
 320 different random seeds is statistically insignificant for these datasets [24]. Additional details of the
 321 experimental setup are provided in Appendix § B. Results with hyperparameter tuning are presented
 322 in Appendix C.

323 **Hidden confounding shift:** Hidden confounding shifts induce observable changes in $\mathbb{P}(\mathbf{X})$,
 324 $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. Table 2 shows nonzero label shift, covariate shift, conditional
 325 covariate shift, and concept shift across the real-world datasets, providing strong empirical evidence
 326 of hidden confounding shift. For the qualitative analysis, we query GPT-4o [1] to list potential hidden

Table 2: Distribution-shift measurements confirm nonzero shift across all four metrics. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples. One-sample t-tests against zero ($\alpha = 0.05$) show that every measure is significantly different from zero (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: The gap *conditional informativeness* – *residual* tracks stronger generalization: larger gaps align with higher average ID and OOD test accuracy over the eight datasets shown in the table on the right.

327 confounders for several benchmark datasets [24, 57]. These results are presented in Appendix § E.
328 For instance, in the *food stamps* dataset, unmeasured factors such as *economic policies* specific to
329 each state may influence both *household income* and *food stamp reciprocity*.

330 **Conditional informativeness vs. accuracy:** Proposition 4.2 makes a clear prediction: under hidden
331 confounding shift, maximizing the difference *conditional informativeness* – *residual* is essential for
332 maximizing predictive information. Results on the first six datasets (that have more than one training
333 environments [57, 24]) listed in Table 2 are shown in Figure 4. Dataset-wise results are in Appendix C.
334 The results confirm the prediction: *conditional informativeness* – *residual* is positively correlated
335 with both ID and OOD accuracy, and XGB achieves both the best performance and the largest value
336 of this gap. As shown in the dataset-wise results in Appendix C, we observe that in many settings,
337 due to hidden confounding shifts, the bars denoted with orange are close to zero because the sum
338 $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ is zero under hidden confounding
339 shift (Proposition 4.2). The relative ID / OOD performance of the methods shown in Figure 1 is
340 consistent with previously reported results in the literature [57, 24].

341 **Informative covariates vs. accuracy:** We next test the mechanism by which informative covariates
342 deliver stronger OOD generalization. To this end, we adopt the covariate partitioning from [57],
343 where all available covariates are grouped into three nested covariate subsets: *causal covariates* $\subseteq$
344 *arguably causal covariates* $\subseteq$ *all covariates* (see Appendix § B). We evaluate how the terms in the
345 decomposition in (1) affect ID and OOD performance when going from one covariate subset to
346 another covariate subset. To this end, we use what we call the *sign consistency metric*, which works
347 as follows. When we move from one covariate set to a larger covariate set, for any term in (1) with a
348 positive coefficient, such as conditional informativeness, we count how often its value rises when
349 accuracy improves. For negatively weighted terms, such as concept shift, we count how often its value
350 decreases when accuracy improves. The metric is the fraction of observations where a term’s change
351 aligns with its *beneficial* direction, thus capturing how reliably it contributes to better generalization.
352 See Appendix § B for the formal definition. Table 3 shows that both conditional informativeness
353 and concept shift exhibit high sign consistency with accuracy gains. While reducing concept shift
354 is crucial, XGB further enhances conditional informativeness compared to other methods, yielding
355 additional performance improvements. Dataset-specific results are presented in Appendix § C.

356 Real-world datasets do not let us verify whether *all available covariates* include *every relevant or*
357 *sufficiently informative* variable. We therefore run a sharper synthetic test of the claim that informative
358 covariates enhance generalization, using the known causal structure: $U \rightarrow X, U \rightarrow Y, X \rightarrow Y$, and

Table 3: Concept shift (CS) shows high sign consistency for all methods, while conditional informativeness (CI) is the decisive term in Proposition 4.2; XGB achieves the strongest CI consistency. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

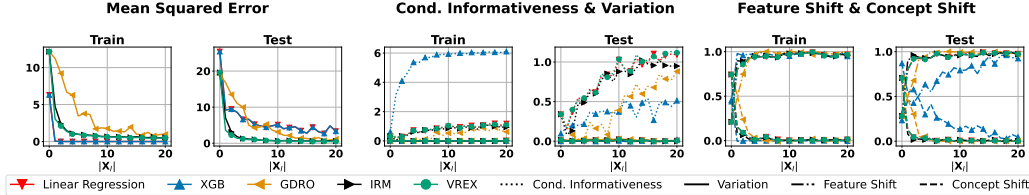


Figure 5: Additional proxy variables $\mathbf{X}_I$ for U that are informative to Y consistently reduce MSE, increase conditional informativeness and feature shift, and reduce concept shift.

359 $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ denotes informative covariates caused by the hidden confounder U . We observe
360 that including $\mathbf{X}_I$ leads to improvements in conditional informativeness and feature shift, while
361 reducing concept shift, as shown in Figure 5. Notably, the variation component remains consistently
362 zero due to the simplicity of the problem. The synthetic data generation process and additional results
363 are provided in Appendix § D.

364 6 Discussion, limitations, and future work

365 We decompose predictive information between model outputs and true outcomes to identify the
366 terms that actually govern OOD generalization (Equation (1)). The analysis shows that hidden
367 confounding shifts, visible through covariate and outcome distribution changes across environments,
368 are unavoidable in realistic settings and cannot be fully eliminated by any finite collection of observed
369 covariates or proxies. Because confounding entangles covariate, label, concept, and feature shifts
370 (inequalities in (2), (3)), methods that handle these shifts independently miss the central difficulty. Our
371 main conclusion is therefore sharp: under hidden confounding shift, learning environment-specific
372 input–output mappings is more powerful than enforcing a single invariant predictor (Equation (4)).
373 This perspective establishes hidden confounding as a first-order design constraint for OOD learning.
374 The next advances should develop algorithms for limited confounder overlap, quantify the cost–
375 accuracy trade-off of collecting informative covariates, manage entangled shifts without untestable
376 proxy assumptions, and model ambiguity from unobserved confounders as a core ingredient of
377 OOD-robust learning.

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656 Appendix

657 A Proofs of propositions

658 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 659 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 660 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

661 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 662 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

663 *Proof.* The causal graph in Figure 3 of the main paper gives the substructure $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$.
 664 In this substructure, $\hat{Y}$ adds no predictive information about Y once either $\phi(\mathbf{X})$ or $\mathbf{X}$ is known,
 665 because the path from $\hat{Y}$ to Y is blocked by $\{\phi(\mathbf{X})\}$ or $\{\mathbf{X}\}$. Hence $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$.
 666 We can now expand $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

667 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

668 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

669 The corresponding decomposition with $\mathbf{X}$ in place of $\phi(\mathbf{X})$ follows immediately by using $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$
 670 and introducing $\mathbf{X}$ into the expansion. $\square$

671 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 672 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 673 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

674 *Proof.* Using the causal graph in Figure 3 of the main paper, we prove the proposition by treating the
 675 two relevant cases separately: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

676 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 677 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

678 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 679 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 680 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

681 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

682 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

683 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

684 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 685 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 686 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

687 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 688 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

689 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

690 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

691 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

692 Because the upper and lower bounds coincide in both cases, the desired equality follows: $I(Y; \hat{Y}) =$
 693 $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

694 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 695 $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

696 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
697 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
698 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
699 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

700 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
701 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
702 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
703 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
704 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

705 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

706 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

707 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
708 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
709 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
710 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
711 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

712 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
713 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
714 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
715 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
716 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

717 B Experimental setup

718 This section specifies the experimental setup used to test the paper’s theoretical claims. The impact
719 of random seeds on the results is statistically insignificant in many settings [24]. For the results
720 in the main paper, we report the mean and standard deviation across five random hyperparameter
721 values (sampled from their respective domains in Table B1) for each dataset-method combination.
722 Since we evaluate models on ID test and OOD test data, to evaluate mutual information terms, we
723 use the nonparametric entropy estimation toolbox [37, 77, 78]. Following [24], we evaluate mutual
724 information using both in-distribution (ID) test data (from training domains) and out-of-distribution
725 (OOD) test data (from test domains).

726 For consistency, we draw 20,000 random sam-
727 ples from each ID and OOD test set when-
728 ever available. Across our experiments, sample
729 size has little effect on the evaluated mutual-
730 information terms. For instance, as shown in
731 Figure B1, the terms in Equation (1) computed
732 for the representations of the XGB model on the
733 *income* dataset stayed relatively consistent as we
734 vary sample size. Features $\phi(\mathbf{X})$ are extracted
735 from the layer preceding the classification head;
736 for XGBoost, we combine output margins and
737 SHAP values from the model. All experiments
738 were conducted on a single NVIDIA RTX A6000 GPU.

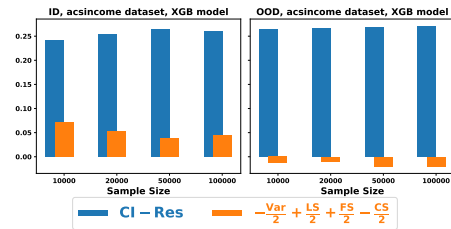


Figure B1: Predictive-information terms remain stable across sample sizes on the *income* dataset.

739 To evaluate how informative covariates affect predictive information, we adopt the categorization
740 of Nastl and Hardt [57] and partition covariates into *causal*, *arguably causal*, and *all covariates*.
741 Causal covariates have clearly established influence on the target, with well-supported directionality
742 and unlikely reverse causation. However, strict reliance on causal variables risks omitting relevant
743 parents due to knowledge gaps. Arguably causal covariates have uncertain causal relationships,
744 meeting at least one of: (1) being known causal, (2) having plausible but potentially bidirectional
745 influence, or (3) likely (but unconfirmed) causal effect. These groups approximate true causal

Table B1: Hyperparameter search space used for the experiments.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]

746 parents based on available knowledge, though relationships may be confounded. Some datasets
 747 additionally contain *anti-causal* covariates where the target likely affects them but not vice versa.
 748 The *all covariates* set includes all observed variables regardless of causal status. The sign consistency
 749 metric introduced in the main paper § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where
 750 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
 751 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

752 Where P is the set of covariate set pairs (s, t) with $t \supset s$. For example, s can be causal covariates
 753 and t can be arguably causal covariates. There is a factor of 2 in $2P$ that accounts for both train and

test data. $\mu_s^{\text{tr/te}}(m)$ is the mean of measure m for covariate set s in training/testing data. $\mathbb{I}(\cdot)$ is the indicator function. $|P| = 3$ for covariate settings $\{\text{causal}, \text{arguably causal}, \text{all}\}$. To get final sign consistency metric value for each m , we average across datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$ where $|D|$ is the number of datasets.

C Additional results on real-world datasets

This section provides additional real-world results that reinforce the main empirical conclusions. Figures C2, C3 show the comparison of the terms: *conditional informativeness* – *residual* and *–variation* / 2 + *label shift* / 2 + *feature shift* / 2 – *concept shift* / 2 for each dataset (their aggregation is in the main paper Figure 4 Left). The key takeaway from these results is that, from Proposition 4.1, the sum of *blue* and *orange* terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of *–variation* / 2 + *label shift* / 2 + *feature shift* / 2 – *concept shift* / 2. This serves as a motivation to maximize conditional informativeness for better OOD generalization under a hidden confounding shift.

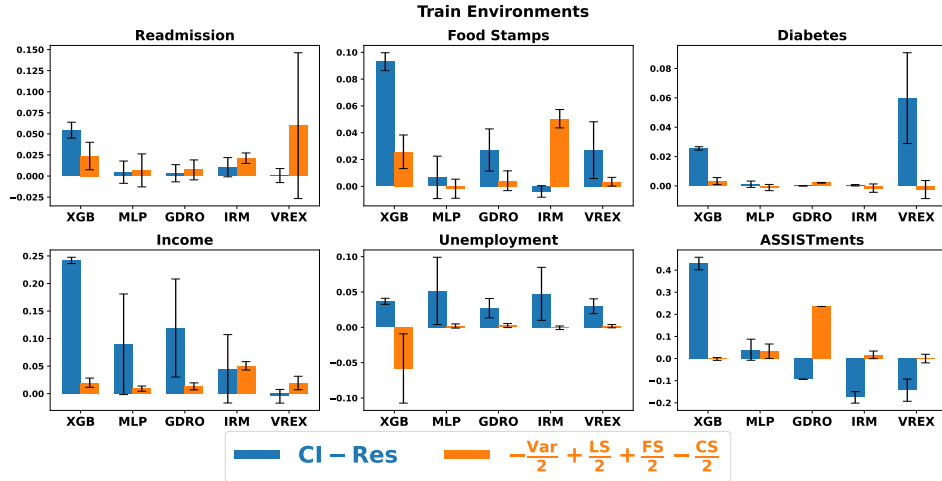


Figure C2: Dataset-specific ID results.

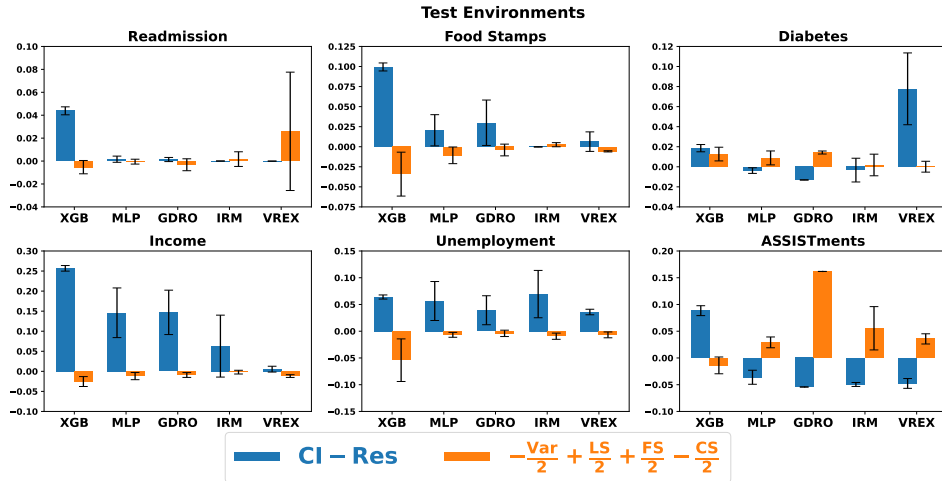


Figure C3: Dataset-specific OOD results.

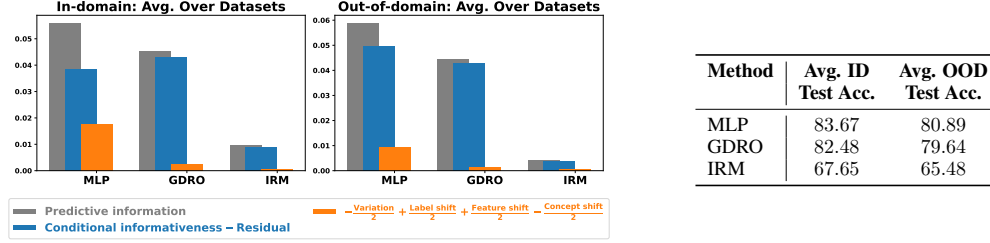


Figure C4: The gap *conditional informativeness* – *residual* in the plots is positively correlated with average ID and OOD test accuracy over the five datasets shown in the table on the right. The contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

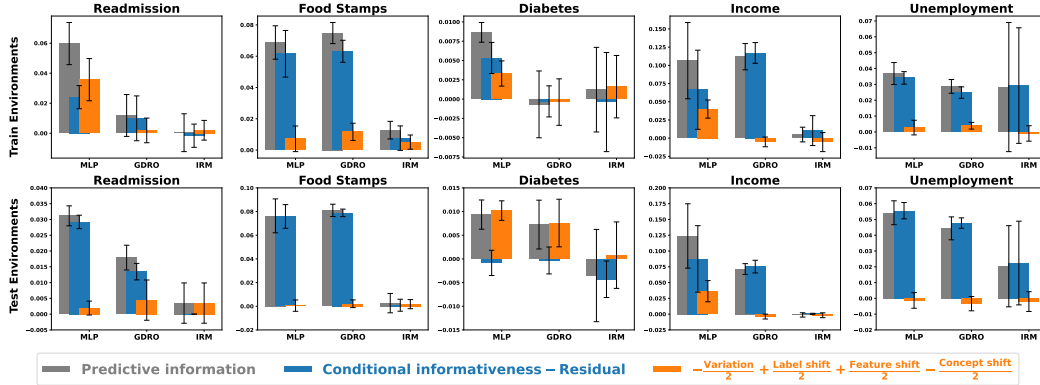


Figure C5: Dataset-specific decomposition of predictive-information terms.

Table C2: Model comparison on ID and OOD test accuracy.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6- C15 further show the comparison of various terms of the information decomposition in (1) for each method and dataset. From these results, we observe that the values plotted in the second column (*variation*) and third column (*feature shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. Similarly, we observe that the values plotted in the fourth column (*label shift*) and fifth column (*concept shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. In almost all settings, the difference *conditional informativeness* - *residual* is the majority contributor for the predictive information.

Hyperparameter tuning: We also perform hyperparameter tuning on model parameters. We run a grid search over hyperparameter values and select the best hyperparameter values based on ID validation accuracy (not OOD validation accuracy). We run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid of hyperparameter values). In this case, to get the mean and standard deviation results for accuracies and mutual information terms, we run

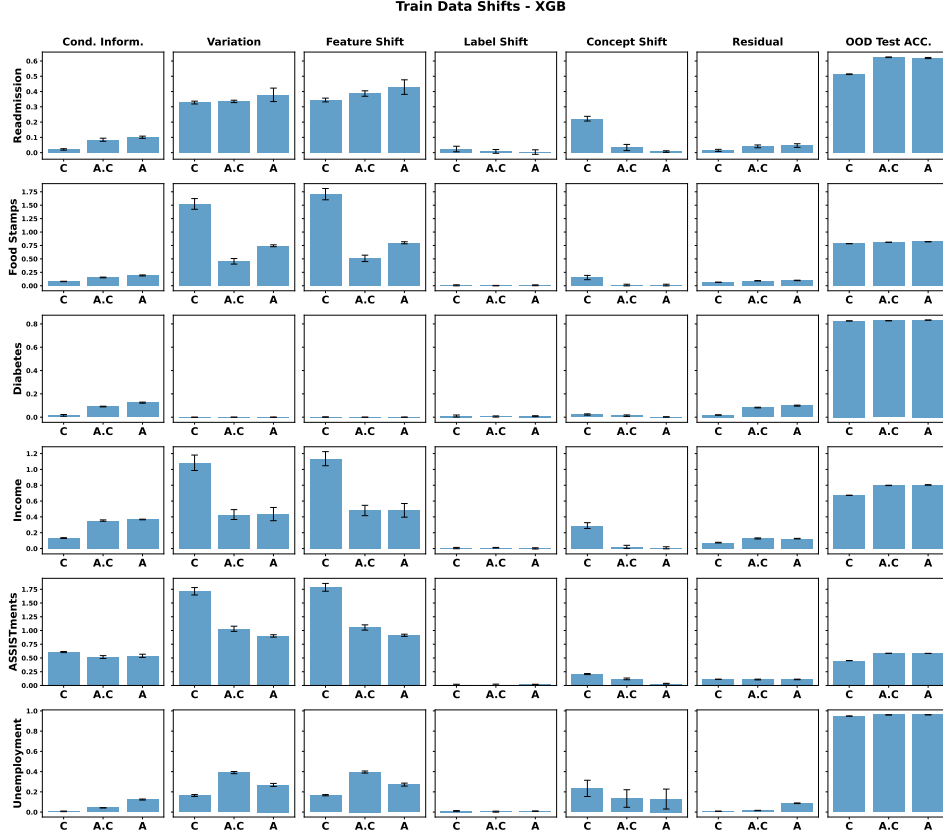


Figure C6: Train-data predictive-information decomposition for XGBoost across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

each experiment for 5 random seeds, even if the effect of random seeds is insignificant in many cases. Results are shown in Figure C4. From Proposition 4.1, the sum of blue and orange terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. Dataset-specific results shown in Figure C5 and Table C2 also show similar trends.

Potential limitations with predictive information decomposition analysis: While predictive information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels, it does not guarantee accuracy. For instance, a binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$) achieves maximal $I(Y; \hat{Y}) = 1$ bit (for binary variables) due to perfect anti-correlation, yet yields 0% accuracy. This occurs because mutual information quantifies *reduction in uncertainty* rather than correctness. Other decomposition terms provide further insight: conditional informativeness $I(\phi(X); Y|E)$ may mask harmful variation $I(\phi(X); E|Y)$, while feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Nevertheless, this decomposition remains valuable for analyzing reasonably performant models, as our experiments demonstrate - it reveals whether predictive information stems from generalizable patterns or environment-specific artifacts, enabling targeted improvements while maintaining interpretability.

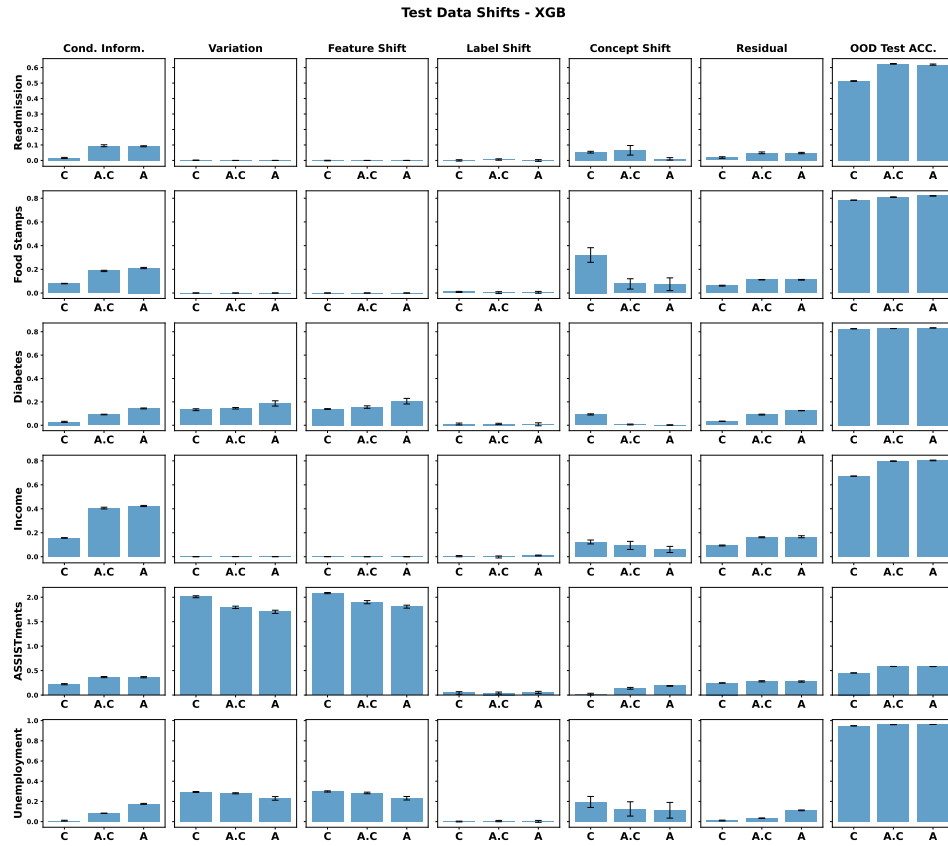


Figure C7: Test-data predictive-information decomposition for XGBoost across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

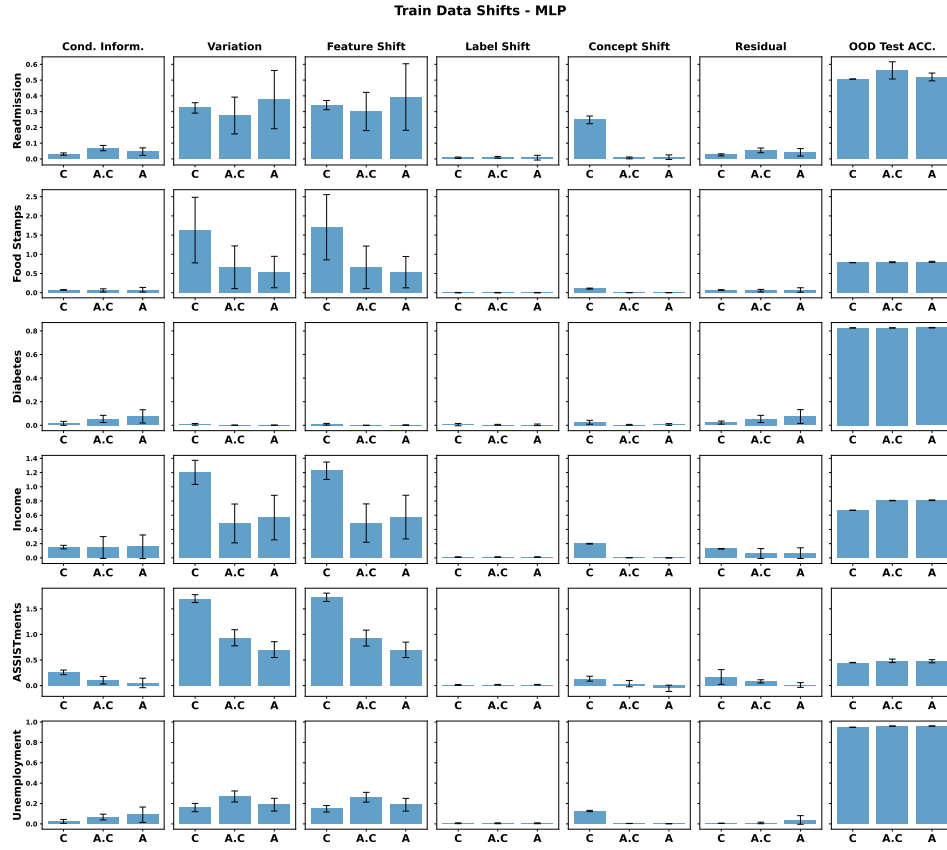


Figure C8: Train-data predictive-information decomposition for MLP across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

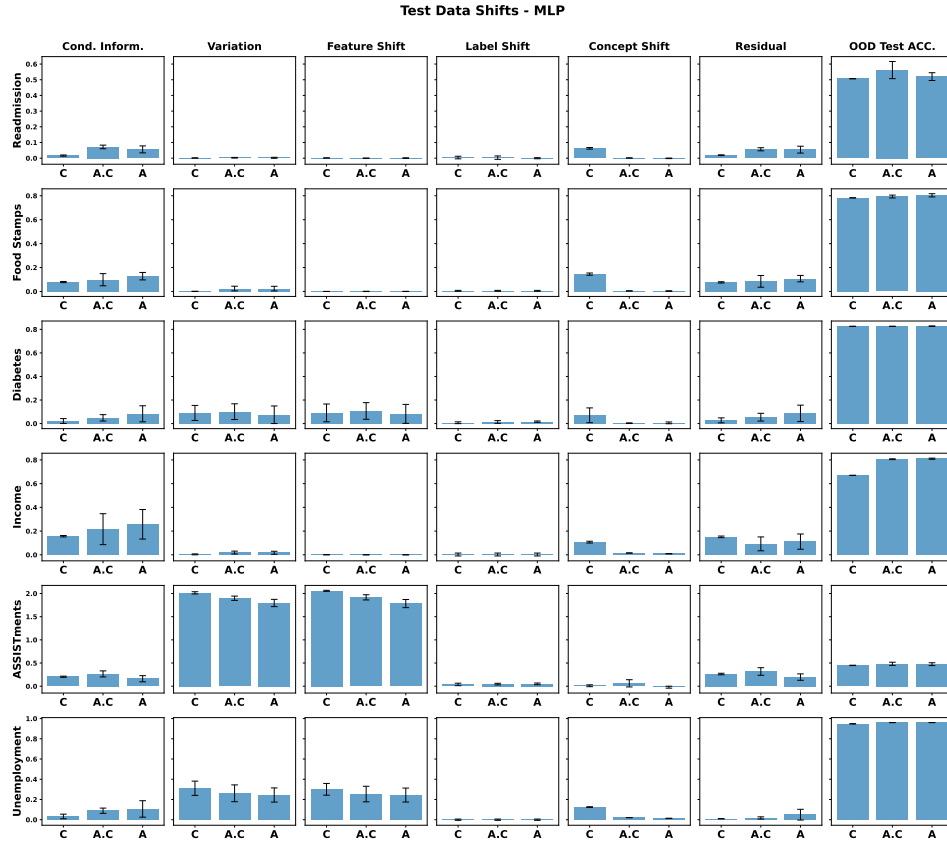


Figure C9: Test-data predictive-information decomposition for MLP across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

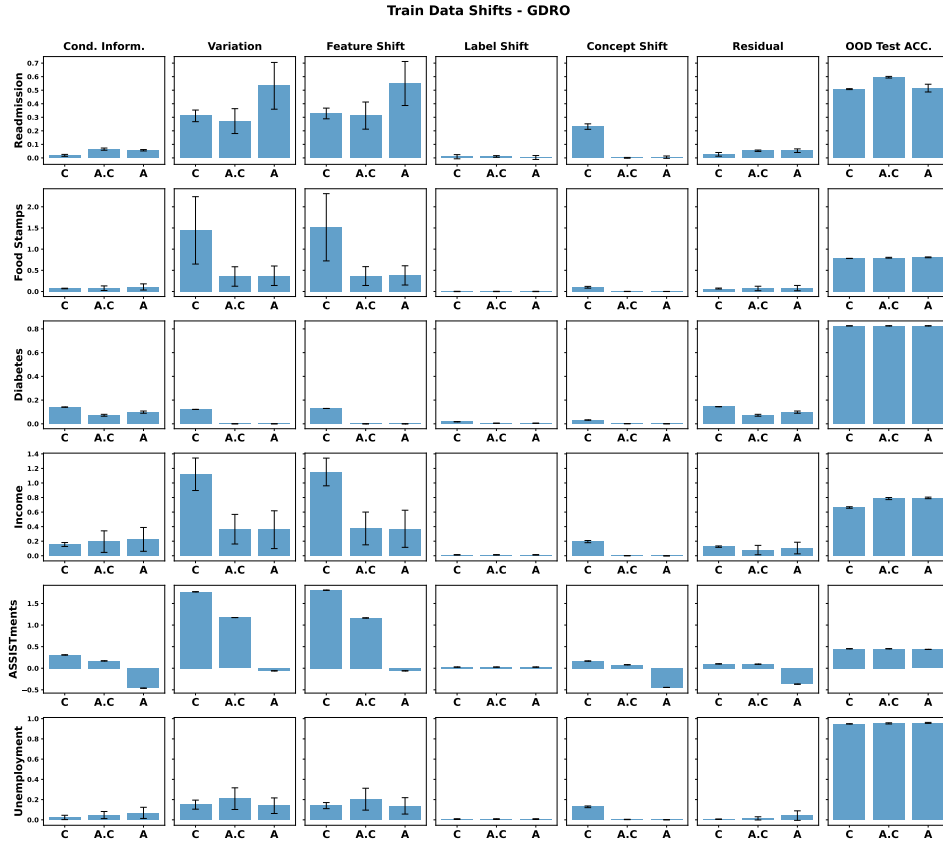


Figure C10: Train-data predictive-information decomposition for GDRO across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

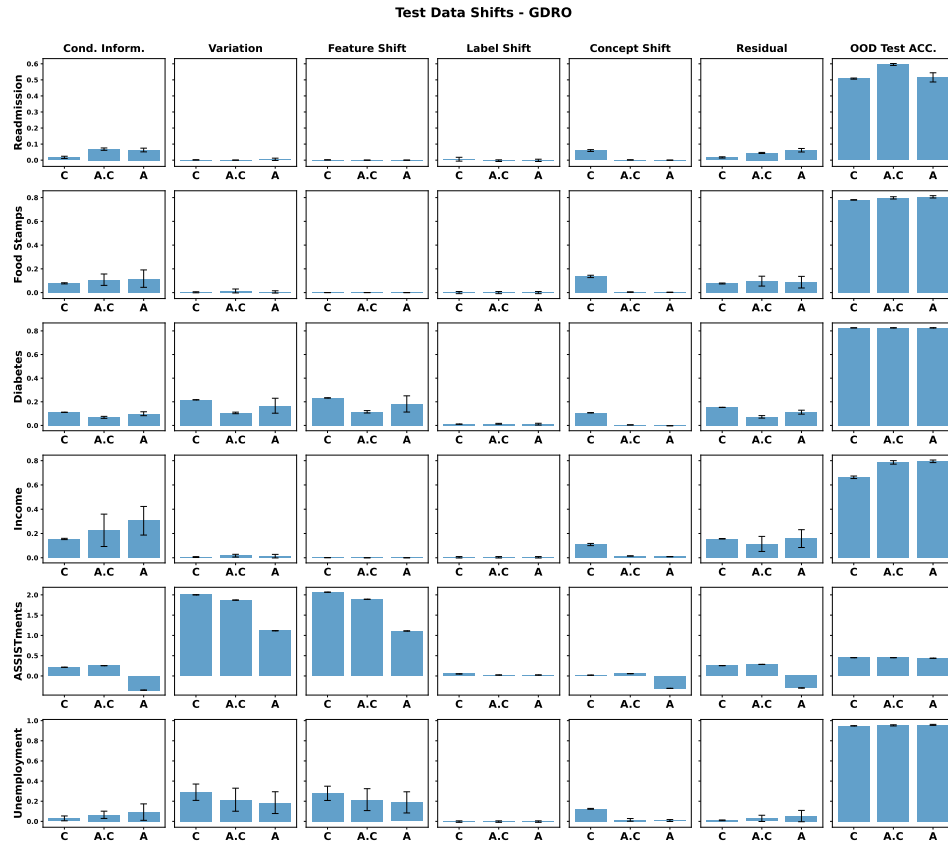


Figure C11: Test-data predictive-information decomposition for GDRO across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

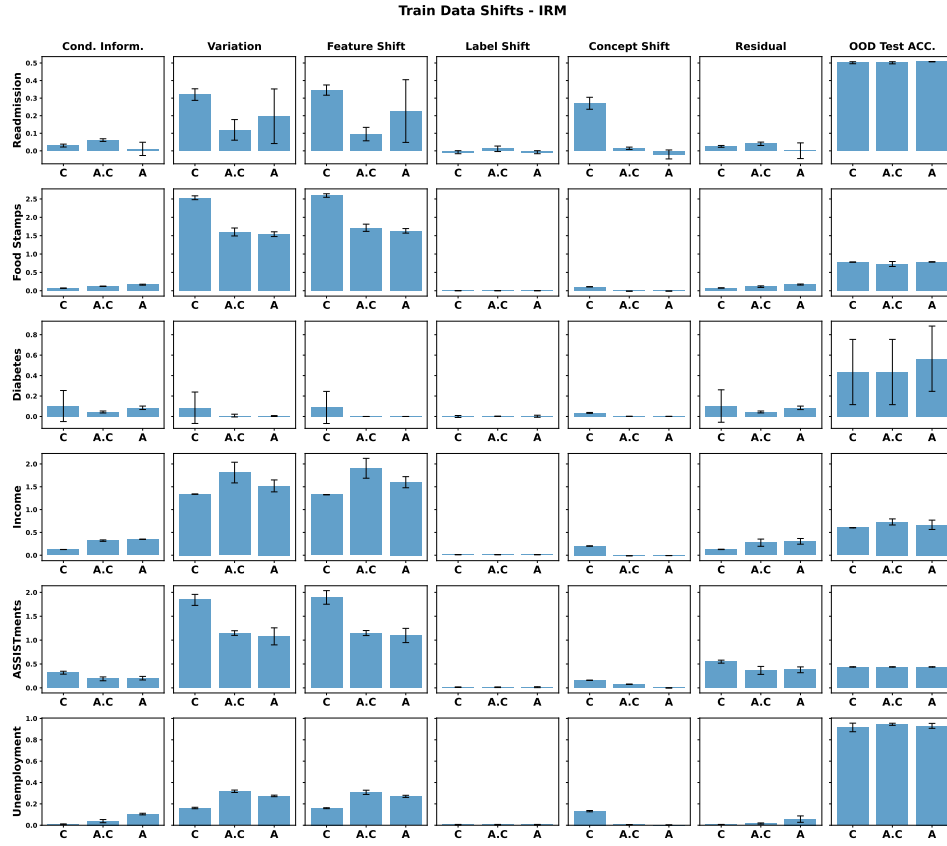


Figure C12: Train-data predictive-information decomposition for IRM across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

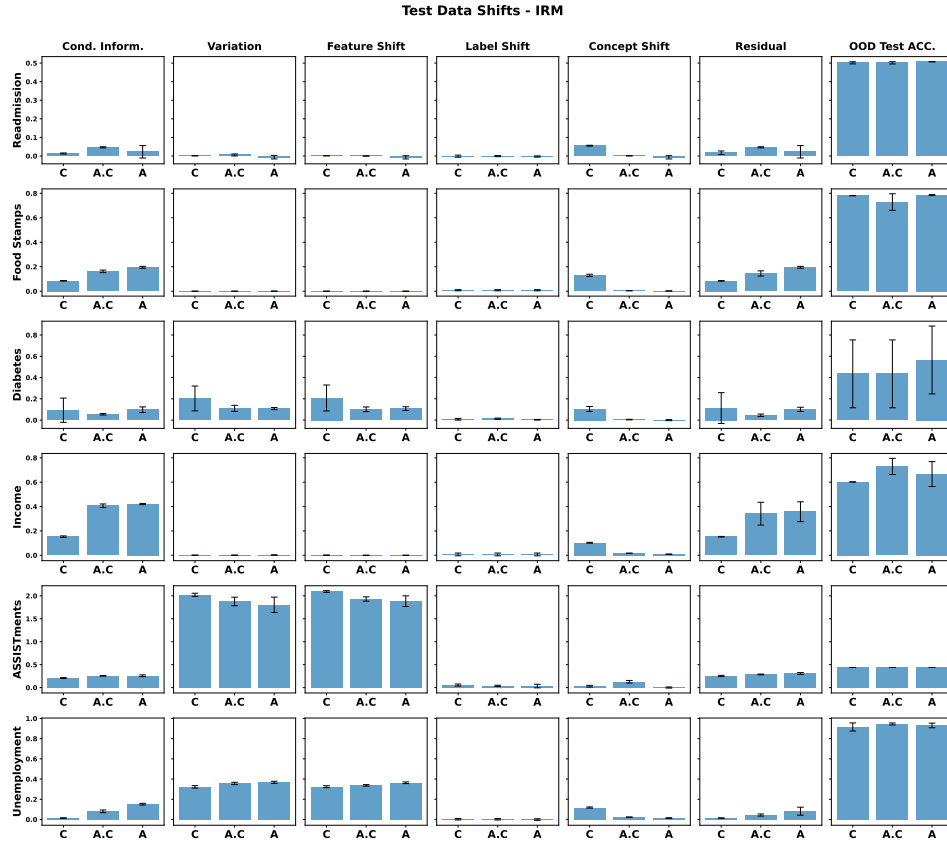


Figure C13: Test-data predictive-information decomposition for IRM across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

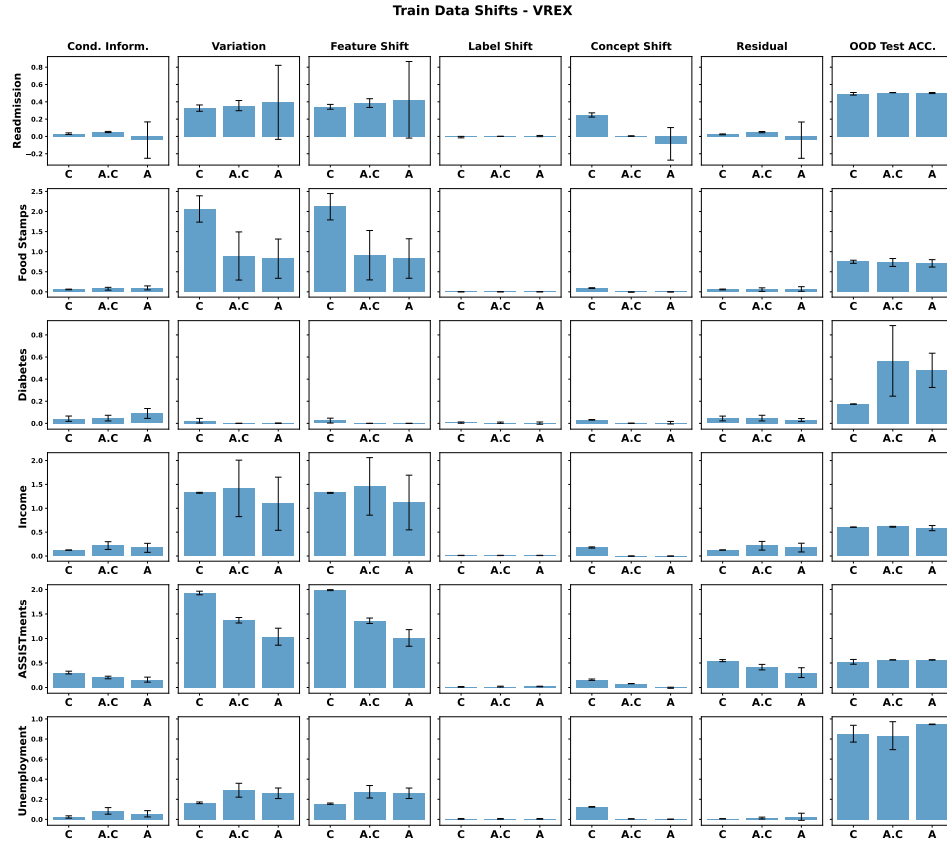


Figure C14: Train-data predictive-information decomposition for VREX across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

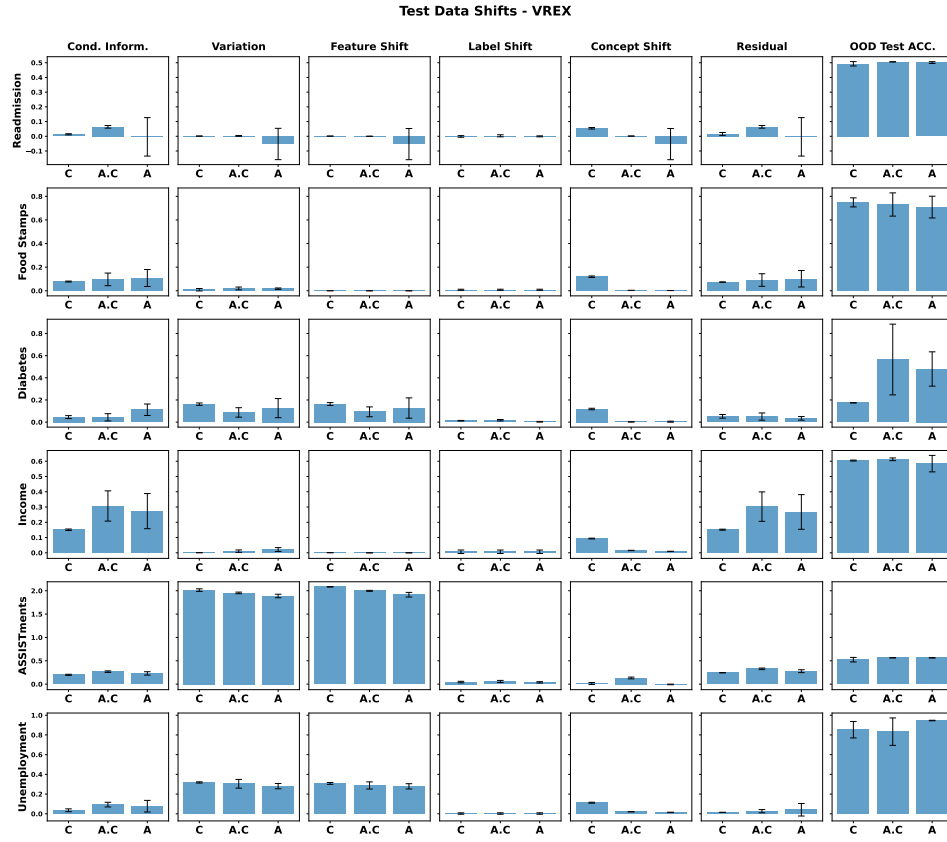


Figure C15: Test-data predictive-information decomposition for VREX across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

D Results on synthetic datasets

The synthetic experiments isolate the mechanisms established by the theory. They test (i) how different informativeness criteria affect generalization, (ii) how confounder-support overlap across environments changes model performance, and (iii) how the balance of causal and anti-causal variables influences accuracy. The data-generating process follows (12) with causal structure $U \rightarrow X$, $U \rightarrow Y$, $U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is an unobserved confounder, X is an observed covariate, Y is the target variable, and $\mathbf{X}_I$ represents additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

We vary the functional forms of f_X , f_i , and f_Y to test the mechanism beyond a single synthetic specification. Domain shifts are induced by systematically changing the environment-specific confounder mean μ_u^e , which gives direct control over the strength of distributional shift across environments.

Informativeness vs. accuracy: We first use linear structural equations, $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$, with $\mu_u^e \in \{-2, 2\}$ for training environments and $\mu_u^e \in \{0, 4\}$ for test environments. Figure 5 in the main paper uses one hidden confounder and $|\mathbf{X}_I| = 20$ informative covariates. Increasing the number of informative covariates from 0 to 20 produces the predicted pattern: lower mean squared error, higher conditional informativeness, higher feature shift, and lower concept shift. We then test two additional informativeness regimes. With one hidden confounder and a single informative covariate ($|\mathbf{X}_I| = 1$), decreasing the noise level $\sigma_i \in \{0, 0.1, \dots, 2.0\}$ from 2.0 to 0.0 again lowers mean squared error, increases conditional informativeness and feature shift, and reduces concept shift (Figure D16). As in the other synthetic settings, the variation term remains zero. Finally, with 20 hidden confounders and 20 corresponding informative covariates under $f_X = 0.3U$, $f_i = 0.2U$, and $f_Y = X - 2U$, adding informative covariates again reduces mean squared error, increases conditional informativeness and feature shift, and reduces concept shift (Figure D17). These experiments confirm that the informativeness mechanism is robust across multiple synthetic criteria.

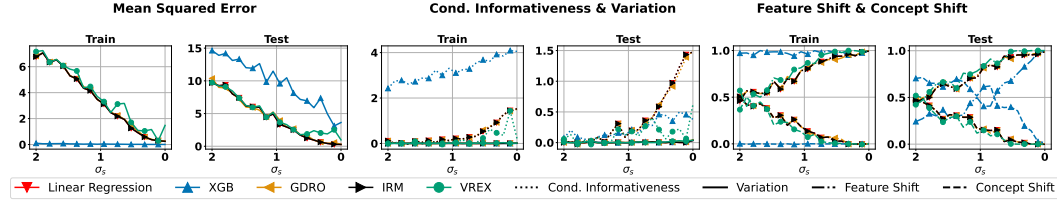


Figure D16: Low-noise proxy variables reduce MSE, increase conditional informativeness and feature shift, and reduce concept shift.

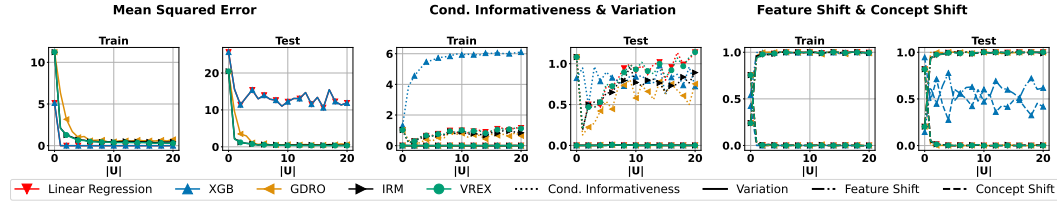


Figure D17: Additional proxy variables $\mathbf{X}_I$ for hidden confounders U that are informative to Y reduce MSE, increase conditional informativeness and feature shift, and reduce concept shift.

Overlapping confounder support vs. accuracy: As established in the main paper, recent work by Prashant et al. [62] proposed an OOD generalization method for hidden confounding shift that assumes test confounder support remains within training support. While our results demonstrated successful learning of correct relationships in synthetic linear data when sufficient information about hidden confounders exists in observed covariates, we now extend this analysis to nonlinear data using MLP and XGB models under both low and high confounding support conditions (Figures D18–D21). In low confounding overlap settings (characterized by distant μ_e between ID and OOD data),

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

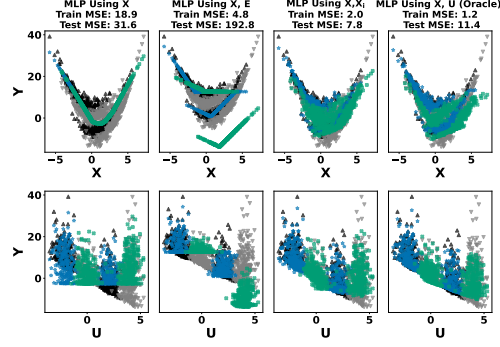


Figure D18: MLP performance in the low-overlap nonlinear setting across feature sets: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

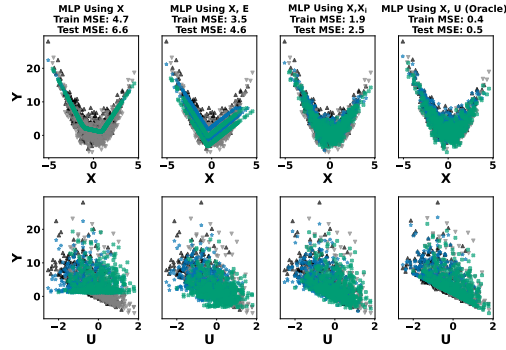


Figure D19: MLP performance in the high-overlap nonlinear setting across feature sets: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

models relying solely on X learn an inadequate global function that fails to distinguish environments. Performance improves when incorporating environment-specific summary statistics of observed covariates, which helps capture environment-specific relationships, but the most significant gains occur only when leveraging additional informative covariates, underscoring their importance. For completeness, we include both oracle model results and U - Y relationship scatter plots to elucidate the learned input-output mappings. These experiments confirm better performance under high confounding overlap as expected, but more importantly, reveal promising results in the low-overlap regime - a previously unstudied scenario that challenges the common support assumption in the literature.

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

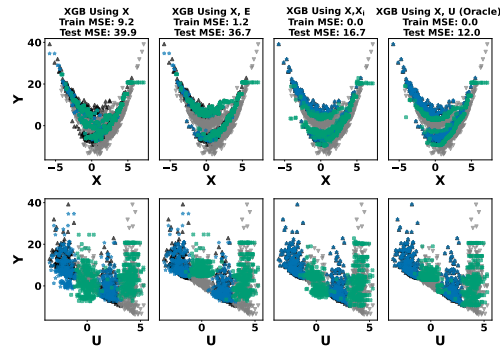


Figure D20: XGB performance in the low-overlap nonlinear setting across feature sets: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

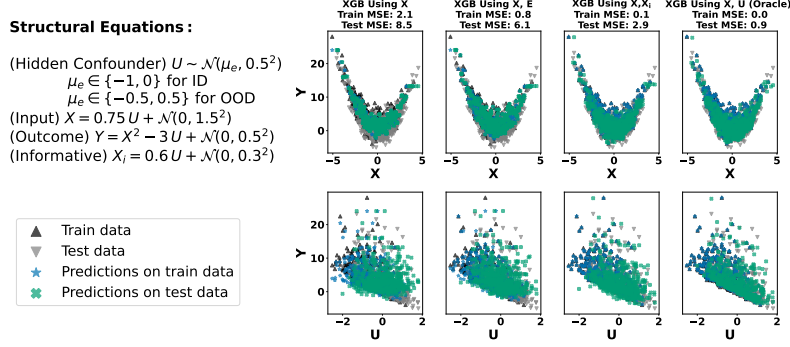


Figure D21: XGB performance in the low-overlap nonlinear setting across feature sets: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Number of causal vs anti-causal variables: Recent benchmarks [24] show that baseline methods (e.g., XGB, MLP) consistently match or outperform OOD generalization techniques (e.g., GDRO, IRM, VREX) in real-world datasets, coinciding with the prevalent causal structure $\mathbf{X} \rightarrow Y$ in tabular data. To test if baseline superiority stems from the causal structure: $\mathbf{X} \rightarrow Y$, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (e.g., $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). Results (Figure D22) show baseline resilience even under the causal structures $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, though no method dominates universally (Figures D23, D24). Baselines achieve superior PMA-OOD scores [24] (fraction of maximum OOD accuracy). We evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) Robust optimization methods (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

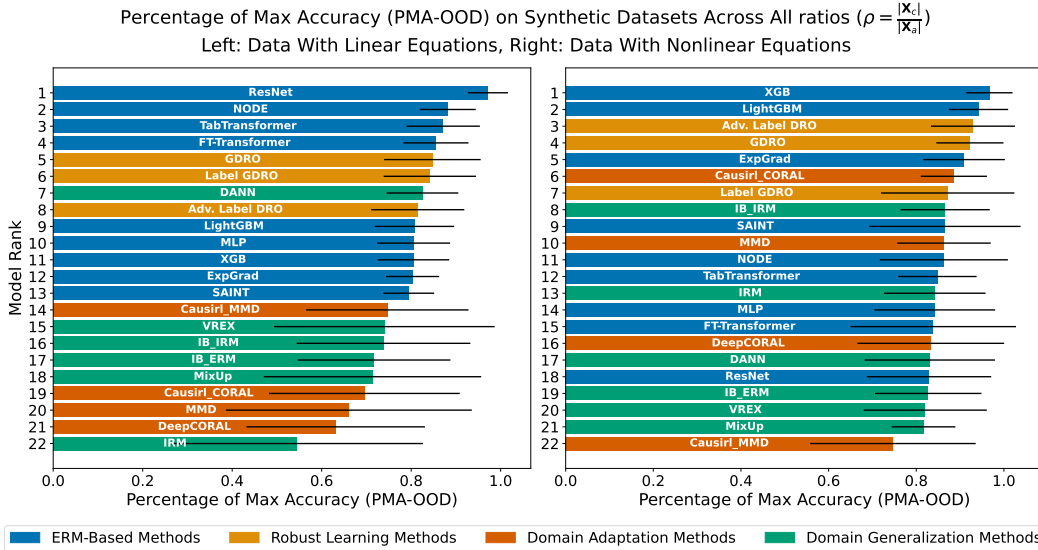


Figure D22: ERM-based methods achieve stronger percentage of maximum OOD (PMA-OOD) test accuracy on the synthetic datasets, consistent with the real-world results of [24].

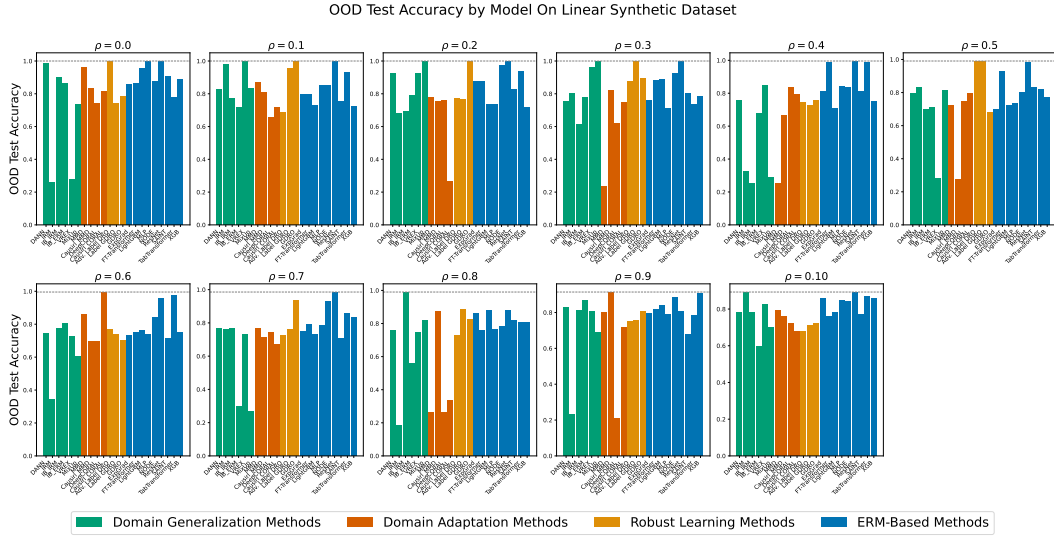


Figure D23: OOD test accuracy in the linear setting for each model across values of ρ .

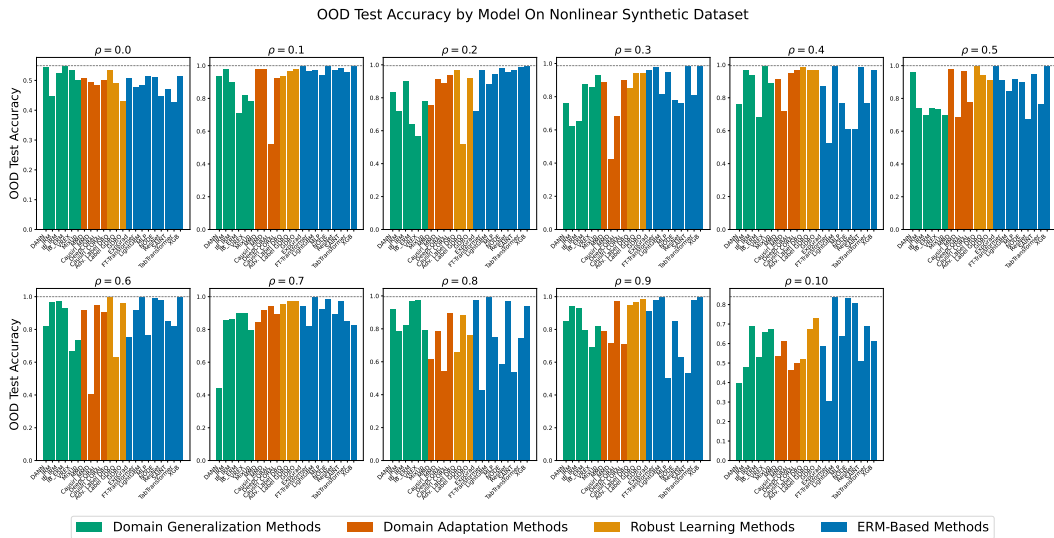


Figure D24: OOD test accuracy in the nonlinear setting for each model across values of ρ

E Qualitative analysis of hidden confounding shift in real-world datasets

This section qualitatively documents hidden confounding shifts in real-world datasets. Tables E3-E18 present potential hidden confounders influencing various observed covariates in several real-world datasets [57]. For additional details regarding these datasets, we refer to [24, 57]. We use GPT4o [1] for this task, and we acknowledge that the list of unobserved confounding variables provided is not exhaustive. This study supports the arguments for the existence and impact of hidden confounding shifts across domains in real-world datasets.

Prompt: We use the prompt below to query GPT4o for plausible hidden confounders.

For a target variable, you will be given lists of causal, arguably causal, anti-causal, and spurious covariates. Also, you will be provided with environment variables such that different datasets are collected in environments induced by these environment variables. Provide possible hidden confounders that cause any of the target, causal, arguably causal, spurious, and anti-causal covariates. That is, the hidden confounders should influence the distribution of their children in different domains. Ensure hidden confounders are not a part of the given variables, and your explanation should include how hidden confounders influence their children in different environments.

Table E3: Candidate hidden-confounder analysis. **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: Candidate hidden-confounder analysis. **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: Candidate hidden-confounder analysis. **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: Candidate hidden-confounder analysis. **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: Candidate hidden-confounder analysis. **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: Candidate hidden-confounder analysis. **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: Candidate hidden-confounder analysis. **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and resources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: Candidate hidden-confounder analysis. **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: Candidate hidden-confounder analysis. **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: Candidate hidden-confounder analysis. **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: Candidate hidden-confounder analysis. **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: Candidate hidden-confounder analysis. **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: Candidate hidden-confounder analysis. **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: Candidate hidden-confounder analysis. **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: Candidate hidden-confounder analysis. **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: Candidate hidden-confounder analysis. **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.